

to compute an instruction context $c_t = f(h'_t, \bar{h})$. When then compute an attentional hidden state $\tilde{h}_t = \tanh(W_c[c_t; h'_t])$, and calculate the predictive distribution over the next action as $a_t = \text{softmax}(\tilde{h}_t)$. Although visual attention has also proved highly beneficial in vision and language problems [60, 33, 3], we leave an investigation of visual attention in Vision-and-Language Navigation to future work.

5.2. Training

We investigate two training regimes, ‘teacher-forcing’ and ‘student-forcing’. In both cases, we use cross entropy loss at each step to maximize the likelihood of the ground-truth target action a_t^* given the previous state-action sequence $\langle s_0, a_0, s_1, a_1, \dots, s_t \rangle$. The target output action a_t^* is always defined as the next action in the ground-truth shortest-path trajectory from the agent’s current pose $s_t = \langle v_t, \psi_t, \theta_t \rangle$ to the target location v^* .

Under the ‘teacher-forcing’ [32] approach, at each step during training the ground-truth target action a_t^* is selected, to be conditioned on for the prediction of later outputs. However, this limits exploration to only states that are in ground-truth shortest-path trajectory, resulting in a changing input distribution between training and testing [45, 32]. To address this limitation, we also investigate ‘student-forcing’. In this approach, at each step the next action is sampled from the agent’s output probability distribution. Student-forcing is equivalent to an online version of DAGGER [45], or the ‘always sampling’ approach in scheduled sampling [8]⁴.

Implementation Details We perform only minimal text pre-processing, converting all sentences to lower case, tokenizing on white space, and filtering words that do not occur at least five times. We set the simulator image resolution to 640×480 with a vertical field of view of 60 degrees. We set the number of hidden units in each LSTM to 512, the size of the input word embedding to 256, and the size of the input action embedding to 32. Embeddings are learned from random initialization. We use dropout of 0.5 on embeddings, CNN features and within the attention model.

As we have discretized the agent’s heading and elevation changes in 30 degree increments, for fast training we extract and pre-cache all CNN feature vectors. We train in PyTorch using the Adam optimizer [28] with weight decay and a batch size of 100. In all cases we train for a fixed number of iterations. As the evaluation is single-shot, at test time we use greedy decoding [44]. Our test set submission is trained on all training and validation data.

⁴Scheduled sampling has been shown to improve performance on tasks for which it is difficult to exactly determine the best next target output a_t^* for an arbitrary preceding sequence (e.g. language generation [8]). However, in our task we can easily determine the shortest trajectory to the goal location from anywhere, and we found in initial experiments that scheduled sampling performed worse than student-forcing (i.e., always sampling).

	Trajectory Length (m)	Navigation Error (m)	Success (%)	Oracle Success (%)
Val Seen:				
SHORTEST	10.19	0.00	100	100
RANDOM	9.58	9.45	15.9	21.4
Teacher-forcing	10.95	8.01	27.1	36.7
Student-forcing	11.33	6.01	38.6	52.9
Val Unseen:				
SHORTEST	9.48	0.00	100	100
RANDOM	9.77	9.23	16.3	22.0
Teacher-forcing	10.67	8.61	19.6	29.1
Student-forcing	8.39	7.81	21.8	28.4
Test (unseen):				
SHORTEST	9.93	0.00	100	100
RANDOM	9.93	9.77	13.2	18.3
Human	11.90	1.61	86.4	90.2
Student-forcing	8.13	7.85	20.4	26.6

Table 1. Average R2R navigation results using evaluation metrics defined in Section 4.4. Our seq-2-seq model trained with student-forcing achieves promising results in previously explored environments (Val Seen). Generalization to previously *unseen* environments (Val Unseen / Test) is far more challenging.

5.3. Additional Baselines

Learning free We report two learning-free baselines which we denote as RANDOM and SHORTEST. The RANDOM agent exploits the characteristics of the dataset by turning to a randomly selected heading, then completing a total of 5 successful `forward` actions (when no `forward` action is available the agent selects `right`). The SHORTEST agent always follows the shortest path to the goal.

Human We quantify human performance by collecting human-generated trajectories for one third of the test set (1,390 instructions) using AMT. The collection procedure is similar to the dataset collection procedure described in Section 4.2, with two major differences. First, workers are provided with navigation instructions. Second, the entire scene environment is freely navigable in first-person by clicking on nearby viewpoints. In effect, workers are provided with the same information received by an agent in the simulator. To ensure a high standard, we paid workers bonuses for stopping within 3m of the true goal location.

6. Results

As illustrated in Table 1, our exploitative RANDOM agent achieves an average success rate of 13.2% on the test set (which appears to be slightly more challenging than the validation sets). In comparison, AMT workers achieve 86.4% success on the test set, illustrating the high quality of the dataset instructions. Nevertheless, people are not infallible when it comes to navigation. For example, in the dataset we occasionally observe some confusion between right and

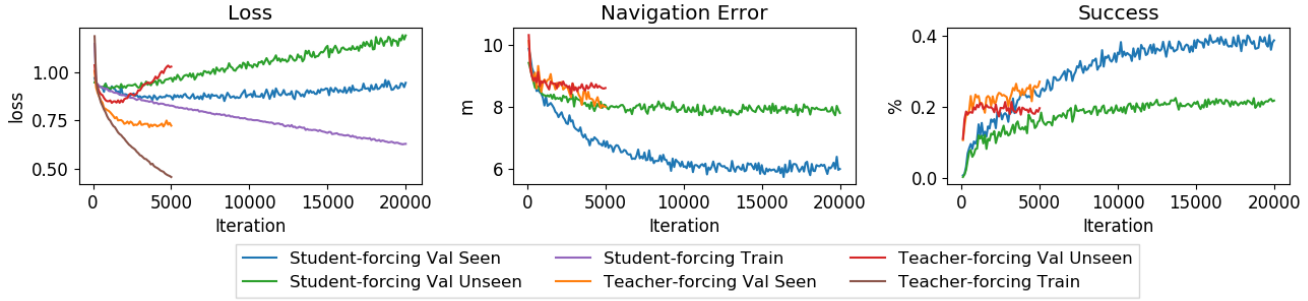


Figure 7. Validation loss, navigation error and success rate during training. Our experiments suggest that neural network approaches can strongly overfit to training environments, even with regularization. This makes generalizing to unseen environments challenging.

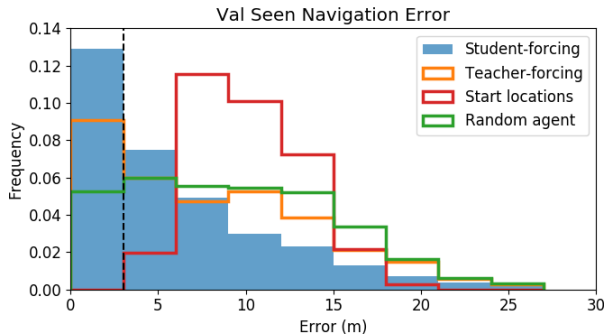


Figure 8. In previously seen environments student-forcing training achieves 38.6% success ($< 3\text{m}$ navigation error).

left (although this is recoverable if the instructions contain enough visually-grounded references). In practice, people also use two additional mechanisms to reduce ambiguity that are not available here, namely gestures and dialog.

With regard to the sequence-to-sequence model, student-forcing is a more effective training regime than teacher-forcing, although it takes longer to train as it explores more of the environment. Both methods improve significantly over the RANDOM baseline, as illustrated in Figure 8. Using the student-forcing approach we establish the first test set leaderboard result achieving a 20.4% success rate.

The most surprising aspect of the results is the significant difference between performance in seen and unseen validation environments (38.6% vs. 21.8% success for student-forcing). To better explain these results, in Figure 7 we plot validation performance during training. Even using strong regularization (dropout and weight decay), performance in unseen environments plateaus quickly, but further training continues to improve performance in the training environments. This suggests that the visual groundings learned may be quite specific to the training environments.

Overall, the results illustrate the significant challenges involved in training agents that can generalize to perform well in previously unseen environments. The techniques

and practices used to optimize performance on existing vision and language datasets are unlikely to be sufficient for models that are expected to operate in new environments.

7. Conclusion and Future Work

Vision-and-Language Navigation (VLN) is important because it represents a significant step towards capabilities critical for practical robotics. To further the investigation of VLN, in this paper we introduced the Matterport3D Simulator. This simulator achieves a unique and desirable trade-off between reproducibility, interactivity, and visual realism. Leveraging these advantages, we collected the Room-to-Room (R2R) dataset. The R2R dataset is the first dataset to evaluate the capability to follow natural language navigation instructions in previously unseen real images at building scale. To explore this task we investigated several baselines and a sequence-to-sequence neural network agent.

From this work we reach three main conclusions. First, VLN is interesting because existing vision and language methods can be successfully applied. Second, the challenge of generalizing to previously *unseen* environments is significant. Third, crowd-sourced reconstructions of *real* locations are a highly-scalable and underutilized resource⁵. The process used to generate R2R is applicable to a host of related vision and language problems, particularly in robotics. We hope that this simulator will benefit the community by providing a visually-realistic framework to investigate VLN and related problems such as navigation instruction generation, embodied visual question answering, human-robot dialog, and domain transfer to real settings.

Acknowledgements This research is supported by a Facebook ParIAI Research Award, an Australian Government Research Training Program (RTP) Scholarship, the Australian Research Council Centre of Excellence for Robotic Vision (project number CE140100016), and the Australian Research Council’s Discovery Projects funding scheme (project DP160102156).

⁵The existing Matterport3D data release constitutes just 90 out of more than 700,000 building scans that have been already been collected [37].

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